1. Aspect 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Concrete manifestation**: The problem provides text descriptions (circular space, three wings, two openings, condition for free airflow) and diagrams (different wing positions, dotted arcs). The solver must correctly understand the meaning of "sealed space" – when openings are too large, the wings cannot simultaneously block the entrance and exit, allowing air to flow straight through. The diagrams also imply the positions of the openings (typically diametrically opposite). This is the basis for building a geometric model.

2. Aspect 5: Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Concrete manifestation**: The problem implicitly involves circles, sectors, equal angles (three wings divide the circle into three 120° sectors), the relationship between arc length and central angle, rotational symmetry, and periodicity. The condition for free airflow translates into a geometric covering problem: given three uniformly spaced radial blades (points) rotating on the circumference, can two fixed arcs (openings) always be blocked by at least one blade? This structure is a typical "points vs. intervals on a circle" covering problem.

3. Aspect 6: Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Concrete manifestation**: Abstract the key mathematical variables from the real situation: radius $R = 100\,\mathrm{cm}$ (diameter 200 cm), number of wings (3), opening arc length L (unknown), central angle $\theta = L/R$, angle between wings (120°), and the central angle between the two openings (inferred as 180°). These variables form the mathematical core of the model.

4. Aspect 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

- **Concrete manifestation**: Simplify the 3D revolving door into a 2D circle problem; ignore wing thickness, treat them as radial lines; ignore aerodynamic details, focus only on geometric blocking; simplify "free airflow" to "there exists a rotation angle where both openings are simultaneously uncovered by any wing"; reduce the continuous rotation problem to an analysis of critical angles.

5. Aspect 2: Identify the key variables in a model

- **Concrete manifestation**: Clearly identify the most critical variable – the opening arc length L (or central angle α); other variables (radius, wing spacing, opening positions) are known or can be assumed. By analyzing the relationships among these variables, determine the maximum allowable value of L.

6. Aspect 3: Select a representation appropriate to the problem context

- **Concrete manifestation**: Use angles (degrees or radians) to represent opening size and wing positions; represent openings as intervals on the circumference (e.g., $[-\alpha, \alpha]$ and $[180^\circ - \alpha, 180^\circ + \alpha]$); represent the three wings as an arithmetic sequence modulo 360° ($\theta, \theta + 120^\circ, \theta + 240^\circ$). This representation is concise and convenient for deriving the covering condition mathematically.

7. Aspect 9: Translate a problem into a standard mathematical representation or algorithm

- **Concrete manifestation**: Convert the original problem into a standard form: find the maximum α (or arc length $L = R\alpha$) such that for every rotation angle θ, at least one of the three wing points falls inside the two opening intervals. Equivalently, find the maximum α such that there is no θ for which all wing points lie in the complement of the openings (the "blocked" arcs). The length of each complement arc is $180^\circ - 2\alpha$, and the wings are spaced by 120°. The critical condition is $180^\circ - 2\alpha = 120^\circ$, giving $\alpha = 30^\circ$, arc length $L = 100 \times \pi/3\,\mathrm{cm}$.

8. Aspect 8: Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **Concrete manifestation**: This problem resembles a classic "revolving door" or "equally spaced points covering arcs on a circle" problem. It relates to concepts like covering a circle, equally spaced points, and interval scheduling. It can be compared to: three equally spaced points – can they simultaneously avoid two symmetric arcs? The solution corresponds to an opening central angle not exceeding 60°, a known geometric fact.

9. Aspects 10/11: Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

- **Concrete manifestation**: Use the arc length formula $L = R \cdot \alpha$, conversion between degrees and radians, inequality $180^\circ - 2\alpha \geq 120^\circ$, etc. Symbolically represent the opening intervals and wing points, set up the covering condition as a mathematical inequality. Optionally draw a diagram of intervals on the circle to aid analysis.

10. Aspect 12: Identify the constraints, assumptions, simplifications in a mathematical model

- **Concrete manifestation**: Explicitly state the assumptions in the model: wings are ideal radial lines (zero thickness); openings are exact arcs; both openings are equal in size and diametrically opposite; free airflow depends only on geometric blocking (no other factors); a wing exactly at the edge of an opening is considered sealed (critical case). These constraints determine the validity range of the model and the precision of the answer.

11. Aspect 1: Select a mathematical description or a representation that describes a problem

- **Concrete manifestation**: Decide to use Euclidean geometry and trigonometry to describe the revolving door structure, rather than algebraic or statistical methods. Choose angular coordinates on the circumference as the basic representation, which provides a clear framework for subsequent analysis.

1. 方面4：阅读、解码并理解陈述、问题、任务、对象或图像以创建模型

- **具体体现**：题目提供了文字描述（圆形空间、三个门翼、两个开口、空气自由流动的条件）和示意图（门翼的不同位置、开口的虚线弧）。解题者需要正确理解“密封空间”的含义——当开口过大时，门翼无法同时遮挡入口和出口，导致空气直通。同时需要从图中推断开口的位置关系（通常为对径分布）。这是建立几何模型的基础。

2. 方面5：识别数学结构（包括规律、关系、模式）

- **具体体现**：题目中隐含了圆、扇形、等分角（三个门翼将圆分成三个120°的扇形）、弧长与圆心角的关系、旋转对称性和周期性。空气自由流动的条件转化为几何覆盖问题：三个均匀分布的径向叶片（点）在圆周上旋转，两个固定弧段（开口）是否总能被至少一个叶片遮挡。这种结构是典型的圆周上的点与区间覆盖模式。

3. 方面6：识别现实问题情境中的数学方面，包括识别重要变量

- **具体体现**：从实际问题中抽象出关键数学变量：圆的半径 $R = 100\,\mathrm{cm}$（直径200 cm）、门翼数量（3个）、开口弧长 L（待求）、开口对应的圆心角 $\theta = L/R$、叶片之间的夹角（120°）、两个开口的中心夹角（推测为180°）。这些变量构成了模型的数学核心。

4. 方面7：简化或分解情境或问题以便进行数学分析

- **具体体现**：将三维旋转门简化为二维圆周上的几何问题；忽略门翼的厚度，视其为径向直线；忽略空气动力学细节，仅考虑几何遮挡；将“空气自由流动”简化为“存在某个旋转角度使得两个开口同时不被任何叶片遮挡”；将连续旋转问题分解为对临界角度的分析。

5. 方面2：识别模型中的关键变量

- **具体体现**：明确最关键的变量是开口弧长 L（或圆心角 α），其他变量如半径、叶片间隔、开口位置均为已知或可假设。通过分析这些变量之间的关系，确定 L 的最大允许值。

6. 方面3：选择适合问题背景的表示

- **具体体现**：使用角度（度或弧度）表示开口大小和叶片位置；使用圆周上的区间表示开口范围（如 $[-\alpha, \alpha]$ 和 $[180^\circ - \alpha, 180^\circ + \alpha]$）；使用模360°的算术序列表示三个叶片的位置（$\theta, \theta + 120^\circ, \theta + 240^\circ$）。这种表示方法简洁且便于进行覆盖条件的数学推导。

7. 方面9：将问题转化为标准数学表示或算法

- **具体体现**：将原问题转化为标准形式：求最大 α（或弧长 $L = R\alpha$），使得对于任意旋转角 θ，三个叶片点中至少有一个落在两个开口区间内。等价地，求最大 α 使得不存在 θ 令所有叶片点都落在两个开口的补集（空白弧段）中。通过分析空白弧段长度 $180^\circ - 2\alpha$ 和叶片间距（120°），得到临界条件 $180^\circ - 2\alpha = 120^\circ$，解得 $\alpha = 30^\circ$，弧长 $L = 100 \times \pi/3\,\mathrm{cm}$。

8. 方面8：识别问题中与已知问题或数学概念、事实或程序相对应的方面

- **具体体现**：该问题类似于经典的“旋转门问题”或“圆上均匀点覆盖弧段”问题，与圆周上的覆盖、等分点、区间调度等概念相关。可类比于：三个等分点能否同时避开两个对称弧段？其解对应于弧段圆心角不超过60°，这是已知的几何结论。

9. 方面10/11：使用数学工具（变量、符号、图表）描述数学结构和关系

- **具体体现**：使用弧长公式 $L = R \cdot \alpha$、角度制与弧度制的转换、不等式 $180^\circ - 2\alpha \geq 120^\circ$ 等数学工具；通过符号化表达开口区间和叶片集合，建立覆盖条件的数学不等式；必要时可画出圆周上的区间图辅助分析。

10. 方面12：识别数学模型中的约束、假设和简化

- **具体体现**：明确模型中的假设：门翼为理想径向直线（无厚度）、开口为精确圆弧、两个开口大小相等且对径分布、空气自由流动仅取决于几何遮挡（无其他因素）、叶片刚好在开口边缘时视为密封（临界条件允许）。这些约束决定了模型的有效范围和答案的精确性。

11. 方面1：选择数学描述或表示来描述问题

- **具体体现**：决定使用欧几里得几何与三角学来描述旋转门结构，而不是代数或统计方法。选择圆周上的角度坐标作为基本表示，这为后续分析提供了清晰的框架。